SPY-SPXW options research

Executive report updated with stock-data comparison · 2026-10-29

Executive summary

The stock-data comparison leaves the research verdict unchanged. No dependable executable edge has been established.

In the separate 2022 legacy sample of 63 expiries, all five stock-source arms produced identical full-cross P&L and no modeled assignment flips. The all-WRDS base-cost mean was +$114.07 per package, with a descriptive 95% interval of -$146.12 to +$367.51.

Removing the modeled overnight stock-price contribution takes that base-cost mean to -$50.46, with an interval of -$99.29 to -$9.93. Overnight stock exposure drives the positive modeled mean.

The nine-expiry 2019 pilot remained negative after all-WRDS replacement, at -$817.49 base-cost mean. The mean changed by +0.0016 USD. These known-data comparisons hold option contracts, quotes and entry cash fixed.

This stock comparison is a separate legacy 2019/2022 fixed-contract experiment, distinct from the 2024/2025 OptionMetrics study. It uses retained original option quotes and contracts. Each position runs from entry to option expiry one calendar day later, then covers stock at the first later scheduled session. The three fills are midpoint, halfway, and full-cross, with full-cross primary. Base and stress means come from dividend-adjusted modeled cost ledgers.

The 2022 descriptive 95% intervals use 10,000 shared ISO year/week cluster draws, seed 20261029, with linear type-7 quantiles. The 2019 pilot has means only.

Recommendation

Keep the strategy in research. Before any live-use claim, test synchronized option and stock quotes under a new frozen protocol, with observed assignment and cover costs.

Earlier OptionMetrics results [1]

Period	n	Expiry intrinsic	Dividend carry	Base cost	Stress cost
2024	195	+$41.09	-$423.53	-$476.37	-$588.40
2025 Jan to Aug	127	+$18.45	+$274.47	+$224.27	+$118.11

Earlier timing-reference means [2]

Period	Reference	n	Expiry	Dividend	Base	Stress
2024	Entry close	195	+$41.09	-$423.53	-$476.37	-$588.40
2024	Prior close	195	-$21.66	-$320.66	-$372.41	-$481.87
2024	Session open	195	+$22.90	-$271.08	-$315.51	-$408.22
2025 Jan-Aug	Entry close	127	+$18.45	+$274.47	+$224.27	+$118.11
2025 Jan-Aug	Prior close	127	-$13.05	+$9.94	-$43.22	-$156.03
2025 Jan-Aug	Session open	127	+$74.00	+$198.72	+$142.01	+$21.23

Sources

[1] Original OptionMetrics theoretical-results report, 2024 and 2025. [2] Earlier timing-reference sensitivity report, entry close, prior close, and session open.

Stock-reference comparison [3]

Full-cross USD package cash P&L; per expiry for a hypothetical 10-short-SPY / 1-long-SPXW package, 100 multipliers. Not account returns. The ledger is a dividend-adjusted modeled cost calculation. Each 2022 estimate carries a descriptive 95% interval; 2019 is a nine-expiry pilot with means only.

Period	Stock reference	n	Expiry intrinsic	Dividend carry	Base cost	Stress cost
2019 pilot	Original stock source	9	-$23.23	-$795.45	-$817.49	-$859.37
2019 pilot	WRDS expiry close	9	-$23.22	-$795.45	-$817.49	-$859.37
2019 pilot	WRDS cover open	9	-$23.23	-$795.44	-$817.49	-$859.37
2019 pilot	WRDS dividends	9	-$23.23	-$795.45	-$817.49	-$859.37
2019 pilot	WRDS all stock inputs	9	-$23.22	-$795.44	-$817.49	-$859.37
2022 Jul-Dec	Original stock source	63	-$15.27 [-$64.92, +$25.49]	+$149.25 [-$111.87, +$404.18]	+$114.07 [-$146.12, +$367.51]	+$42.28 [-$219.74, +$296.64]
2022 Jul-Dec	WRDS expiry close	63	-$15.27 [-$64.92, +$25.49]	+$149.25 [-$111.87, +$404.18]	+$114.07 [-$146.12, +$367.51]	+$42.28 [-$219.74, +$296.64]
2022 Jul-Dec	WRDS cover open	63	-$15.27 [-$64.92, +$25.49]	+$149.25 [-$111.87, +$404.18]	+$114.07 [-$146.12, +$367.51]	+$42.28 [-$219.74, +$296.64]
2022 Jul-Dec	WRDS dividends	63	-$15.27 [-$64.92, +$25.49]	+$149.25 [-$111.87, +$404.18]	+$114.07 [-$146.12, +$367.51]	+$42.28 [-$219.74, +$296.64]
2022 Jul-Dec	WRDS all stock inputs	63	-$15.27 [-$64.92, +$25.49]	+$149.25 [-$111.87, +$404.18]	+$114.07 [-$146.12, +$367.51]	+$42.28 [-$219.74, +$296.64]

2022 paired changes versus original-source baseline

Full-cross arm-minus-baseline paired mean and descriptive 95% interval, fixed n=63. Source [4] records independent numerical validation.

Stock reference	Expiry intrinsic	Dividend carry	Base cost	Stress cost
WRDS expiry close	+$0.00 [+$0.00, +$0.00]	+$0.00 [+$0.00, +$0.00]	+$0.00 [+$0.00, +$0.00]	+$0.00 [+$0.00, +$0.00]
WRDS cover open	+$0.00 [+$0.00, +$0.00]	+$0.00 [+$0.00, +$0.00]	+$0.00 [+$0.00, +$0.00]	+$0.00 [+$0.00, +$0.00]
WRDS dividends	+$0.00 [+$0.00, +$0.00]	+$0.00 [+$0.00, +$0.00]	+$0.00 [+$0.00, +$0.00]	+$0.00 [+$0.00, +$0.00]
WRDS all stock inputs	+$0.00 [+$0.00, +$0.00]	+$0.00 [+$0.00, +$0.00]	+$0.00 [+$0.00, +$0.00]	+$0.00 [+$0.00, +$0.00]

Sensitivity findings and limits

Single-source diagnostic changes use full-cross base-cost means versus the original-source baseline. They are not additive components of the all-WRDS arm.

Period	Stock reference	Base-cost mean change
2019 pilot	WRDS expiry close	less than $0.0001 in magnitude
2019 pilot	WRDS cover open	+$0.0016
2019 pilot	WRDS dividends	+$0.00
2022 Jul-Dec	WRDS expiry close	+$0.00
2022 Jul-Dec	WRDS cover open	+$0.00
2022 Jul-Dec	WRDS dividends	+$0.00

Full-cross unique expiry assignment flips [3]

Period	Stock reference	Unique flips	n expiries
2019 pilot	WRDS expiry close	0	9
2022 Jul-Dec	WRDS expiry close	0	63
2019 pilot	WRDS cover open	0	9
2022 Jul-Dec	WRDS cover open	0	63
2019 pilot	WRDS dividends	0	9
2022 Jul-Dec	WRDS dividends	0	63
2019 pilot	WRDS all stock inputs	0	9
2022 Jul-Dec	WRDS all stock inputs	0	63

Limits

The comparison changes stock reference inputs. Option quotes, contracts, entry cash, and settlement stay unchanged. This legacy comparison retains original option quotes; old/new stock-source clock semantics are unresolved. Stock-source sensitivity does not establish which provider is correct. Actual or early assignment, fills, capacity, margin, and funding are unobserved. The earlier 2024/2025 OptionMetrics study pairs 15:59 ET BBO with later close references, so it does not establish execution. The periods are known data, not untouched holdouts.

Dividend check: no fixed interval spanned the 2022-09-16 ex-date. One selected expiry was 2022-12-15 with scheduled cover on the 2022-12-16 ex-date, but it was unassigned, so eligible assigned carry was zero. The modeled dividend difference was not rounded away.

Assignment-flip rows overlap across fills. The table counts distinct expiry values in stored full-cross records only. It does not sum across fills or multiply expiry counts by three.

Status and provenance

Run status: FIVE_ARM_HISTORICAL_SENSITIVITY_EVALUATED. Manifest git SHA: ac0b05e3bce2d6c4a176bb49a392f47ae2912306. Validation [4]: PASS on the same SHA, 141403 checks, max absolute error 3.183231456205249e-12. Test counts describe code behavior, not report effectiveness. Full input, command, source inventory, and receipts are in Markdown.

Sources

[3] Five-arm evaluator artifacts, including summaries, uncertainty, and stored flips. [4] Independent checker receipt and test log.